SCALPTrade LLC C++ Coding Challenge

The purpose of this challenge is to assess your coding skills and style. Your code is expected to be reasonably efficient. You are not allowed to use any third-party library, only features available in the C++14/11 standard. Please submit code that can be compiled in Linux without the use of third-party tools.

The program will be given the following input parameters provided as command line arguments:

1. Symbol to be traded
2. Order Side 'B' or 'S'
3. Maximum order size
4. VWAP window time period in seconds.
5. TCP market data ip address
6. TCP market data port
7. TCP order connection address
8. TCP order connection port

Your task is to create a program as follows:

- Use a TCP socket to connect to the market data IP address and port, and listen for market data packets in the format specified below.
- Use a TCP socket to connect to the order connection IP address and port, and send an order in the data format specified below.
- VWAP is Volume Weighted Average Price. For the purpose of this program, you need to calculate the VWAP price using the time interval that is passed in the input:

  $$\text{VWAP} = \left(\sum_{i=1}^{n} P_i \cdot Q_i\right) / \left(\sum_{i=1}^{n} Q_i\right)$$

- P is Trade Price, Q is a trade quantity in the VWAP formula above.
- At the end of each VWAP window time period, the program needs to calculate VWAP Price, and use it to determine whether to send an order or not in response to each of the subsequent incoming quotes. The first trade marks the beginning of a window.
- The last calculated VWAP price should be used while values are being accumulated for the calculation of VWAP for the next VWAP window.
- For a Buy order, "better price" is defined as the market data quote price being less than the calculated VWAP price. For a sell order, "better price" is defined as a market data quote price greater than the calculated VWAP price.
- Send an order for every market quote whose price is better than the calculated VWAP price assuming a valid VWAP price has been calculated.
- Orders can not be sent until the initial VWAP price is calculated (minimum one full period)
- Order size should match the current quote size up to the maximum specified by the max order size input parameter
- For simplicity, each run of the algo will only buy or sell.
- Also, for simplicity, there is no maximum number of orders to be sent or position limit.

Sample application input parameters provided as command line argument:

Myapp IBM B 100 30 127.0.0.1 14000 127.0.0.1 15000

- Symbol = IBM
- side=B
- max order size = 100
- VWAP window time period = 30 seconds
- TCP market data IP = 127.0.0.1
- TCP market data port = 14000
- TCP order IP = 127.0.0.1
- TCP order port = 15000

Sample market data

Trades (without header, and not in binary)
IBM 1580152659000000000 100 13900
IBM 1580152659000000001 50 13899
IBM 1580152659000000002 30 13898
IBM 1580152659000000003 98 13884
Etc

Quotes (without header, and not in binary)
IBM 1580152659000000000 5 13897 9 13898
IBM 1580152659000000001 11 13896 10 13897
IBM 1580152659000000002 24 13895 41 13897
Etc

Submissions should include the following:

- A tar file of the code with instructions and a makefile
- A brief README describing the design, any assumptions used and known issues.
- Any test cases used to verify performance

Message Protocol

You are responsible for defining the message protocol. The only requirements are that the messages maintain little endian byte order and have the following fields:

Quote

Field	Description
Symbol	Symbol string; at most 6 characters
Timestamp	Nanoseconds since epoch
Bid Quantity	Number of contracts
Bid Price	Price in pennies
Ask Quantity	Number of contracts
Ask Price	Price in pennies

Trade

Field	Description
Symbol	Symbol string; at most 6 characters
Timestamp	Nanoseconds since epoch
Quantity	Number of contracts
Price	Price in pennies

Order

Field	Description
Symbol	Symbol string; at most 6 characters
Timestamp	Nanoseconds since epoch
Side	BUY or SELL
Quantity	Number of contracts
Price	Price in pennies

Guidelines:

- Use C++ best practices as appropriate.
- You will be judged by the structure of your code, latency and performance of the code, in addition to how well the solution works.
- You may not use any third-party libraries or packages
- If you have any questions, you can email us.